BZ350 Molecular and General Genetics
Final Exam (May 13, 2025)

NAME ____________________________

This exam is 5 pages long. You are responsible for verifying that your exam copy contains all pages. Please write your name on the top of all pages.

SECTION A: Multiple choice questions (3 pts each; Total 33 pts): Circle the **most appropriate** answer for each question.

1. Which of the following features in **prokaryotic mRNAs** is important for the recognition of the **initiation codon by the 30S ribosome during translation**?
   A) TATA box
   B) Shine-Dalgarno box
   C) Cap
   D) CAAT box
   E) promoter

2. Glucocorticoids (one type of steroid hormone) **induce the expression of specific genes** by
   A. binding and inactivating a transcriptional activator
   B. binding and activating a transcriptional repressor
   C. binding and relocating a transcription factor from cytoplasm to nucleus
   D. binding and inactivating a transcriptional repressor

3. Which of the following mutation(s) results in **constitutive expression** of *lac* operon (i.e., high expression in the presence or absence of lactose)?
   A) Lac I^-
   B) O^c
   C) I^s
   D) Lac I^- or O^c
   E) CRP^-

4. Which of these statements **does NOT** describe a characteristic of the genetic code?
   A) The code is degenerate
   B) The genetic code is universal i.e., nearly all living organisms use the same genetic code.
   C) Each codon specifies only one amino acid.
   D) Each amino acid is coded by only one codon
   E) Codons are nonoverlapping

5. In a DNA cloning experiment, you inserted a fragment of human DNA into a *LacZ* (β-galactosidase) gene in a vector with an ampicillin resistance gene, transformed it into *E.coli* cells, and grew them on plates containing ampicillin and X-Gal. **Which of the following colonies would contain the recombinant plasmid** (i.e., plasmid with human DNA)?
   A) The blue colonies
   B) The white colonies
   C) All of the colonies
   D) None of the colonies

6. Which of the following proteins is a **positive regulator** of gene expression in **bacteria**?
   A) Protein coded by *LacI*
   B) Protein coded by *TrypR*
   C) cyclic AMP receptor protein (CRP)
   D) GAL80

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7. In Trp R^- *E.coli* cells (i.e., these cells do not produce the repressor), in the presence of a **high amount of tryptophan** in the medium the attenuator region forms which of the following stem-loop(s)?

A. 1-2 stem-loop
B. 2-3 stem-loop
C. 3-4 stem-loop
D. 1-2 and 3-4 stem-loops

8. Nucleotides that are used to **terminate DNA synthesis** in Sanger's DNA sequencing method do not contain a

A) 1'-OH
B) 2'-OH
C) 3'-OH
D) 4'-OH
E) 5'-OH

9. A mutation introduced **a termination codon in the middle of an exon (coding region)** of a human gene. What would happen to the expression of this gene?

A) This gene gets transcribed but is not spliced.
B) This gene gets transcribed, and processed properly to produce mature RNA, which produces a truncated (shorter) protein.
C) This gene does not get transcribed.
D) This gene gets transcribed but is not capped and polyadenylated.

10. The lack of methylation of which of the following results in maternal imprinting of *igf2* gene expression?

A) promoter
B) enhancer
C) insulator
D) terminator
E) TATA box

11. Which of the following small non-coding RNAs suppress the expression of transposons?

A) miRNAs
B) piRNAs
C) siRNAs
D) All of them
E) None of them

SECTION B: Fill-in questions. Fill in the blank with a word or phrase that **best completes** the sentence. **(3 pts each; Total 12)**

12. ______________________________________Proteins such as the TrpR or lacI repressor that undergo changes in conformation (shape) when bound to other molecules are called _____________ proteins.

13. ____________________________________ DNA regulatory sequences that activate gene expression but are located far away from a gene in the upstream or downstream of a gene are called _____.

14. ____ **Leucine zipper domain** found in many transcription factors is necessary to ______ two proteins.

15.________________________________Proteins that can activate gene expression **without binding the DNA** are called _______________.

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SECTION C: SHORT ANSWER QUESTIONS (TOTAL 52 pts). Answer all questions in the space provided.

16. For each of the following *E. coli* haploid or partial diploids containing the *lac* operon alleles listed, predict the expression (inducible, constitutive, no expression, or basal expression) of β-galactosidase and permease in the absence and presence of lactose. Explain your prediction. (**9 pts**).

A. $I^s\ O^+\ Z^+\ Y^+$

B. $I^-\ O^+\ Z^-\ Y^-\ /\ I^+\ O^c\ Z^+\ Y^-$

C. $I^s\ O^+\ Z^+\ Y^+\ /\ I^-\ O^+\ Z^+\ Y^-$

17. What are microRNAs (miRNAs)? How are they produced? Describe two mechanisms by which they regulate gene expression. (**10 pts**)

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18. If you grow bacterial (E. coli) cells in the presence of both glucose and lactose, what will happen to **lac operon** gene expression, and why? (**6 pts**)

19. What would happen to the expression of galactose inducible genes in the following yeast mutants in the presence and absence of galactose? Explain your prediction. (**9 pts**)

i) $GAL4^-$ and $GAL80^-$

ii) $GAL4^-$

iii) $GAL80^-$

20. In polymerase chain reaction (PCR) the amount of the DNA is doubled in each cycle of amplification. Describe the three steps involved in one cycle of amplification**.** List any two applications of PCR. **(8 pts)**

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21. For the following *E. coli* strains indicate the effect of the genotype on the expression of the *trpE* and *trpC* genes in the presence and absence of tryptophan. Provide an explanation for your prediction. **(6 pts)**

a. $R^{-}\ P^{+}\ O^{+}\ att^{+}\ trpE^{+}\ trpC^{+}$

b. $R^{+}\ P^{-}\ O^{+}\ att^{+}\ trpE^{-}\ trpC^{-} /\ R^{-}\ P^{+}\ O^{+}\ att^{+}\ trpE^{+} trpC^{+}$

22. What is a reporter gene in molecular genetics and explain the utility of reporter genes? Name two commonly used reporter genes. (**4 pts**).

Section D: Indicate whether the following statements are either TRUE or FALSE. (1 pt each)

____________ The primary structure of proteins, i.e., its amino acid sequence, determines the tertiary and quaternary structure of proteins.

_______________ Although there are no operons in **eukaryotic** cells, the expression of multiple genes can be regulated by a single enhancer.

_______________ Methylation of cytosine in DNA, a common epigenetic modification, generally results in **more accessible chromatin (open chromatin)**.

HONOR PLEDGES: You have the option of signing the following statements.
I have not given, received, or used any unauthorized assistance.

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END OF EXAM